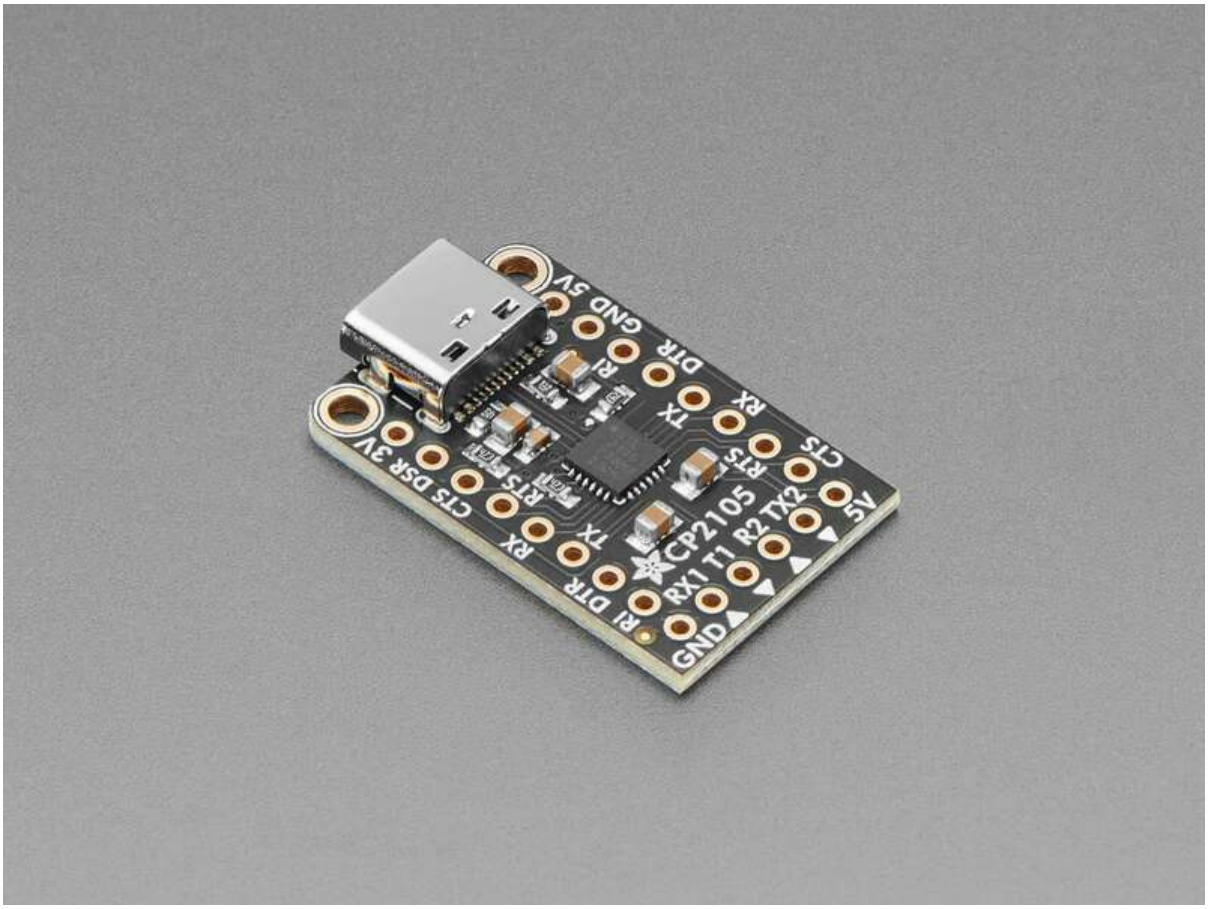
Adafruit CP2105 Friend - USB to Dual UART Serial Converters

Created by Liz Clark



https://learn.adafruit.com/adafruit-cp2105-friend-usb-to-dual-uart-serial-converters

Last updated on 2026-10-19 02:40:02 PM UTC

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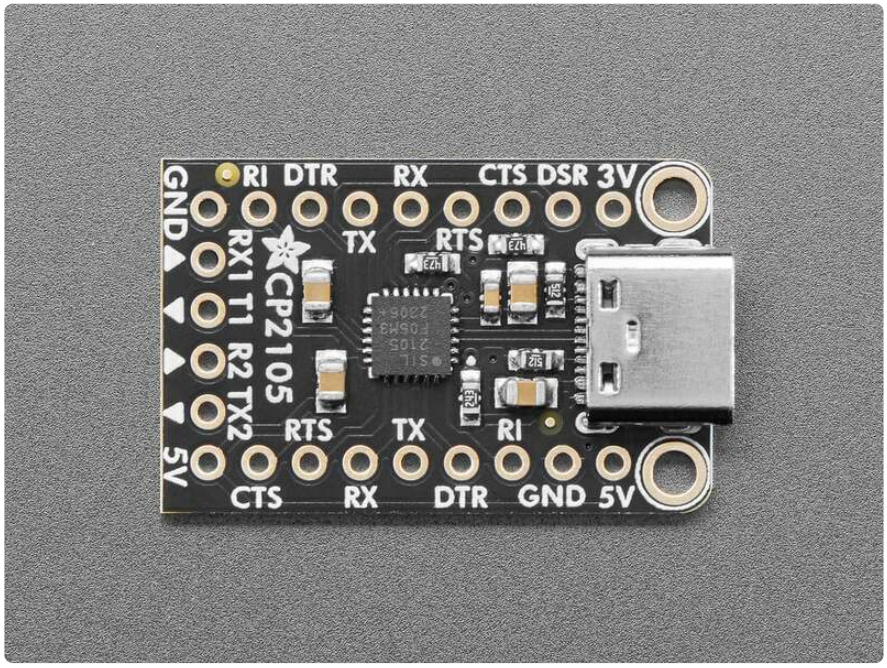
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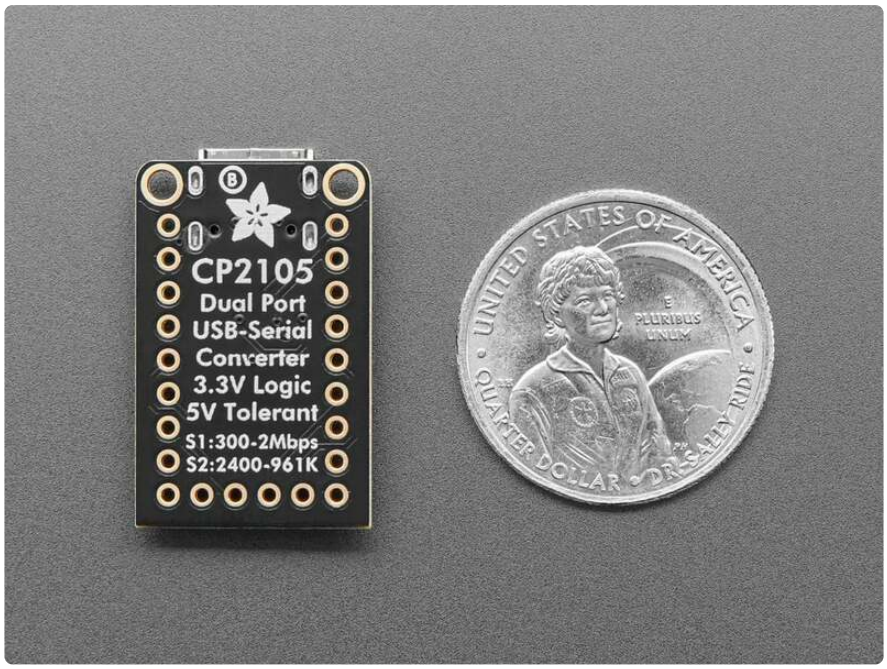
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Overview



On our first Pentium 120 tower (purchased at PCs for Everyone on Thorndike St in Cambridge Mass ()) came with two DE-9 serial ports on the back: COM1 and COM2. Perfect for connecting a mouse and maybe a modem. Nowadays, long gone are the days of parallel ports and serial ports. Now the USB port reigns supreme! But USB is hard, and you just want to transfer your every-day serial data from two serial ports to a single USB port without the complexity of a hub. What now? Enter the **Adafruit CP2105 Friend**!



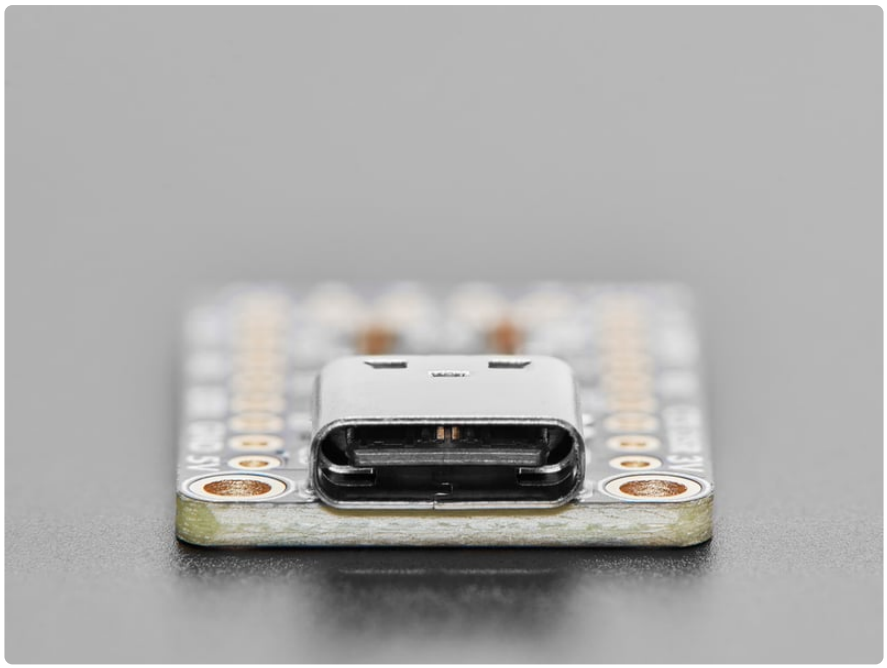
The CP2105 is very similar to the CP2102N (http://adafru.it/5335). Except, instead of having a single serial UART port, it has two. One is 'standard' and one is 'enhanced' - you'll get both enumerated when plugging in this device, without any hub required!

The **Enhanced** port can do:

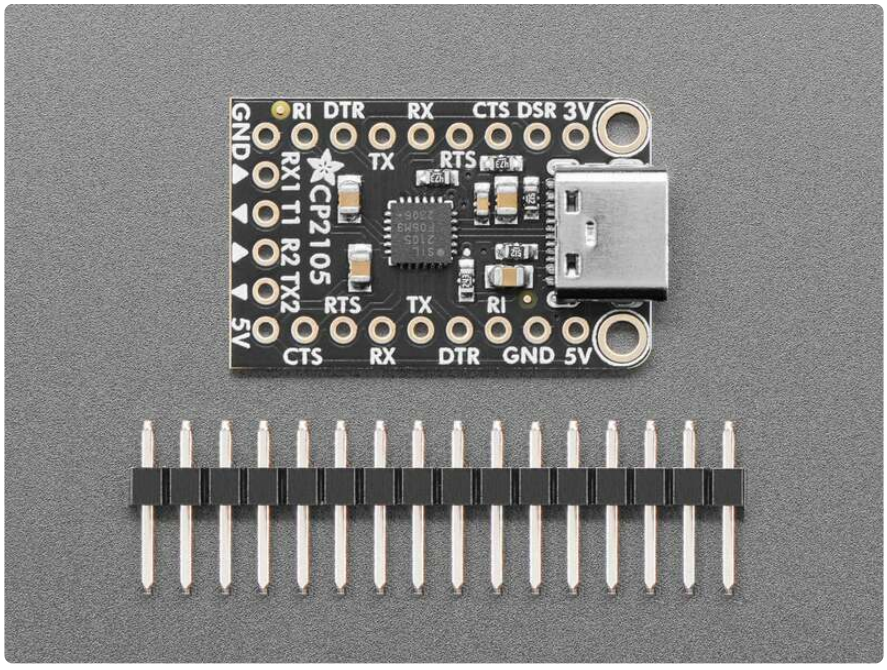
- Data formats supported:
  - Data bits: 5, 6, 7, and 8
  - Stop bits: 1, 1.5, and 2
  - Parity: odd, even, mark, space, no parity
- Baud rates: 300 bps to 2.0 Mbps
- 320 Byte receive and transmit buffers
- Modem control signals: CTS, RTS, DTR, DSR and RI

The **Standard** port can do:

- Data format: 8 data bits, 1 Stop bit
  Parity: Even, Odd, No parity
- Baud rates: 2400 bps to 921600 bps
- 288 Byte receive and transmit buffers
- Modem control signals: CTS, RTS, DTR, DSR and RI



Like the CP2102N breakout, we have USB C connector and the same overall shape, but there are some significant changes. The pins are going to be totally different because we have two UARTs, one on either 'side' of the board. The left side is the Enhanced port, right side is the Standard port. We don't have extra GPIO pads and there are no indicator LEDs. Also the bottom row is a 'minimal' output of just the two RX/TX pads and power pins.



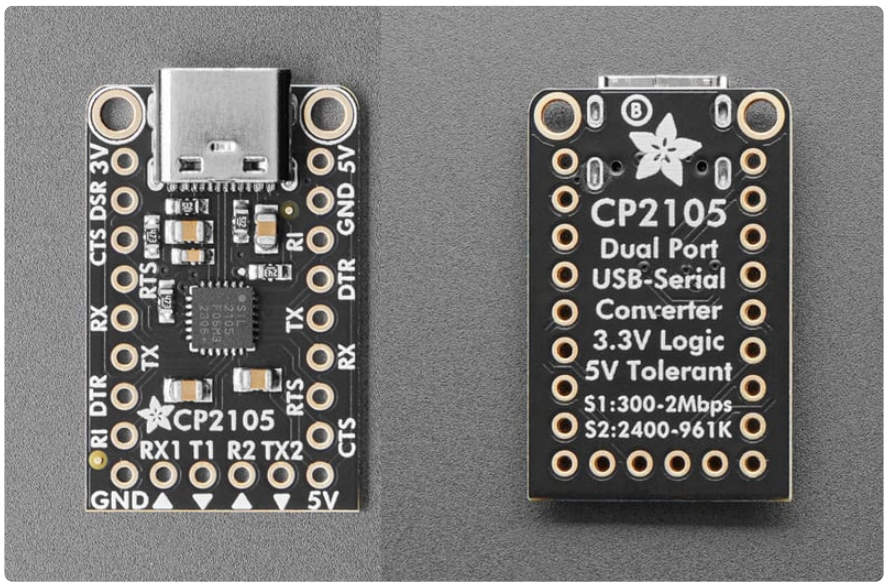
We like the CP210x series because it has better driver support than the CH340 and can do very high speeds, and variable speeds without stability issues. Compared to

the FT232RL and FT231X, the CP210x has the same capabilities or better, at a great price!

Each order comes with a fully assembled and tested board. We give you some pin header strip. Solder it in to plug it into a breadboard and get access to all the pins.

For Linux you won't need a driver. For Windows, it will automatically grab the driver from Windows Update. For Mac OS X you can check out SiLabs driver page for the latest and greatest ().

Pinouts



Power Pins

- **5V** - These are the 5V power pins. They receive 5V output from USB and can supply ~100mA peak.
- **3V** - This is the 3.3V power pin. It is the output from the 3.3V regulator and can supply ~100mA peak.
- **GND** - Common ground for power and logic.

Enhanced Port Pins

The enhanced port pins are broken out on the left side of the board.

- **RX/RX1** - UART data receive pin. This pin receives the serial input. This pin is also available along the bottom edge of the board as **RX1**.

- **TX/T1** - UART data transmit pin. This pin sends the serial output. This pin is also available along the bottom edge of the board as **T1**.
- **DSR** - Data Set Ready control input. This pin is active low.
- **CTS** - Clear To Send control input. This pin is active low.
- **RTS** - Ready To Send control output. This pin is active low.
- **DTR** - Data Terminal Ready control output. This pin is active low.
- **RI** - Ring Indicator control input. This pin is active low.

The enhanced port can handle baud rates from 300 bps to 2.0 Mbps and 320 Byte receive and transmit buffers. It supports the following data formats:

- Data bits: 5, 6, 7, and 8
- Stop bits: 1, 1.5, and 2
- Parity: odd, even, mark, space, no parity

Standard Port Pins

The standard port pins are broken out on the right side of the board.

- **RX/R2** - UART data receive pin. This pin receives the serial input. This pin is also available along the bottom edge of the board as **R2**.
- **TX/TX2** - UART data transmit pin. This pin sends the serial output. This pin is also available along the bottom edge of the board as **TX2**.
- **DSR** - Data Set Ready control input. This pin is active low.
- **CTS** - Clear To Send control input. This pin is active low.
- **RTS** - Ready To Send control output. This pin is active low.
- **DTR** - Data Terminal Ready control output. This pin is active low.
- **RI** - Ring Indicator control input. This pin is active low.

The standard port can handle baud rates from 2400 bps to 921600 bps and 288 Byte receive and transmit buffers. It supports the following data formats:

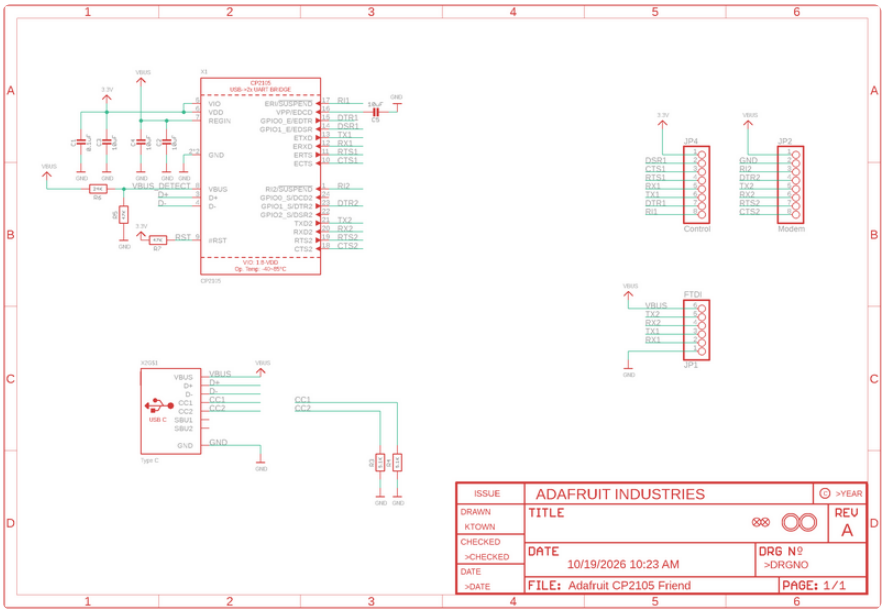
- 8 data bits
- 1 stop bit
- Parity: even, odd, no parity

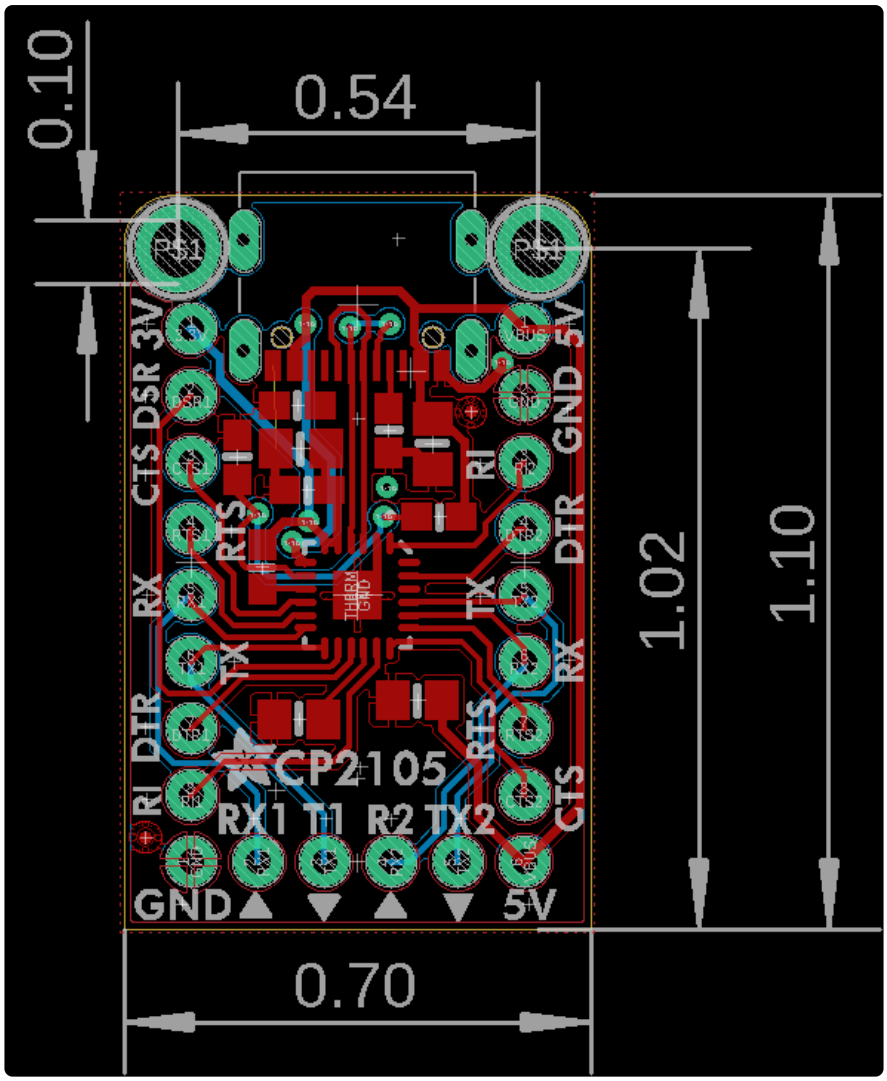
Downloads

Files

- Windows and Mac Drivers Download ()
- CP2105 Datasheet ()
- AN197 Serial Communications Guide for the CP210x ()
- EagleCAD PCB Files on GitHub ()
- Fritzing object in the Adafruit Fritzing Library ()

Schematic and Fab Print




0.10
0.54
1.02
1.10
0.70
3V
DSR
CTS
RTS
RX
TX
DTR
RI
5V
GND
RI
DTR
TX
RX
RTS
CTS
TX
RI
CP2105
RX1 T1 R2 TX2
GND
5V